MZ.510 Assignment 1

Due: Monday, October 26, 2026 by 23:59.

Please email your report to M. Cagri Gurbuz (mgurbuz@zlc.edu.es)

Problem 1 (15 POINTS):

Three refineries with maximum daily capacities of 6, 5, and 6 million gallons of oil supply three distribution areas with daily demands of 4, 8 and 7 million gallons. Oil is transported to the three distribution areas through a network of pipes. The transportation cost is $1 per 100 gallons per mile. The mileage table below shows that refinery 1 is not connected to distribution area 3:

		Distribution Area		
		1	**2**	**3**
Refinery	**1**	120	180	-
	2	300	100	80
	3	200	250	120

Additionally, suppose that distribution area 1 must receive all its demand, and any shortage at areas 2 and 3 will result in a penalty of $5 per gallon.

Formulate the problem as a transportation model and solve it to minimize the costs while meeting all constraints.

Problem 2 (15 POINTS):

Three parts are assembled to produce one unit of a particular product. Each unit of this final product requires 3 units of Part 1, 1 unit of Part 2, and 2 units of Part 3. The production can be done in two departments (Department A and Department B). The following table summarizes the required time (in hours) to produce one unit of each part in different departments, along with the total time available in Departments A and B:

	Department A	Department B
Part 1	0.8 hr	0.6hr
Part 2	0.3 hr	0.4hr
Part 3	0.5 hr	0.6 hr
Available time	150 hrs	150 hrs

Formulate a linear program and solve it to maximize the weekly production of the final product.

Problem 3 (20 POINTS):

Transeast Airlines fly planes on the route "LA- Houston-NYC- Miami-LA". The length in mile of each trip is as follows:

Trip segment	Miles
LA - Houston	1500
Houston-NYC	1700
NYC-Miami	1300
Miami-LA	2700

The price of aviation fuel at each airport varies significantly and is in table below:

City	Price per gallon ATF in cents
LA	88
Houston	15
NYC	105
Miami	95

The plane's fuel tank can hold at most 12000 gallons. To allow for the possibility of circling over a busy landing site we require that the ending fuel level for each leg be at least 600 gallons. The number of gallons used per mile of the flight can be estimated by (a consulting firm built a regression model)

$$\text{gallons used per mile} = 1 + \left(\frac{\text{average fuel level during the leg}}{2000} \right)$$

To simplify the formulation, assume that the average fuel level on any leg of the flight is

$$\left(\frac{\text{fuel level at the start of leg} + \text{fuel level at the end of leg}}{2} \right)$$

Formulate and solve an LP model which can be used to minimize the fuel cost for the entire LA- Houston-NYC- Miami-LA route (assume that the plane has the same route for a long enough planning horizon – that it repeats the same route over and over). Find the optimal solution and discuss insights observed from the model output.

Problem 4 (15 POINTS):

Below are the supply, demand and shipping cost data for a specialty steel company:

Location	Capacity (tons)
A. Charlotte	90
B. Raleigh	50
C. Lexington	80
D. Danville	60

Location	Demand (tons)
1. Richmond	120
2. Winston-Salem	100
3. Durham	110

Shipping Costs	To		
From	**1**	**2**	**3**
A	$70	$100	$50
B	$120	$90	$40
C	$70	$30	$110
D	$90	$50	$70

1. Formulate the problem as an LP and determine an optimal plan to meet the demand as much as possible at the three locations.
2. If demand in Durham were to decrease by 10 units, would the solution change?
3. The company board has decided to add capacity at the Lexington plant. Present a quantitative argument in favor of or against the decision using sensitivity analysis from your model. Clearly show how much will be the gain (if any) per unit capacity increase in Lexington and compare with adding capacity at other locations.

Problem 5 (15 POINTS):

The production manager for the Classic Boat Corporation must determine how many Model A21 boats and Model B32 boats to produce over the next three quarters. The company has a beginning inventory of 100 Model A21 boats and 80 Model B32 boats. The demands for the two models, as well as the capacities, of the three quarters are given in the following table. The capacity is in the number of boats regardless of model. Management has specified that the ending inventory for quarter 3 must be at least 300 Model A21 and 200 Model B32 boats, and that the demand must be met 100% and on time.

Quarter	Demand		Capacity
	A21	B32	
1	1200	380	4000
2	2500	1500	3500
3	1800	1200	2000

The production costs for the first quarter are $8,000 per Model 21 boat and $10,000 per Model B32 boat. These costs are expected to increase by 10% each quarter because of increases in labor and material costs. Inventory holding cost is 5% of the production cost for each boat held in inventory at the end of the quarter. Please help the production manager to formulate and solve a linear programming model to determine the production for each of the three quarters so as to meet the demand and minimize the total cost.

Problem 6 (10 POINTS):

Texxon Oil has three active oils in a Texas oil field. Well capacities are as below:

Well	Capacity TBD (Thousand Barrels per Day)
1	93
2	88
3	95

The company operates five refineries around the gulf coast with refining output (assumed fixed for the purpose of this exercise) as below:

Refinery	1	2	3	4	5
Prod. TBD	30	57	48	91	48

There are three pump stations which are used to transport oil from the wells to the refineries. Oil can flow from any well to any pump station and likewise from any pump station to any refinery. The pipelines used are operated by a third party pipeline company which charges $/TBD according to pipeline segment used as per the schedule below:

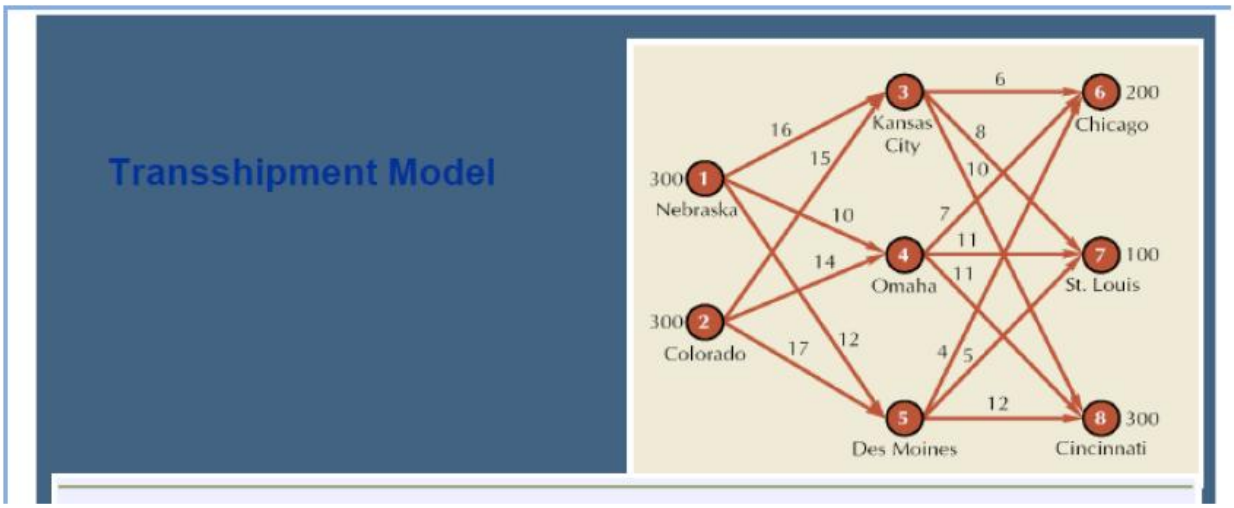
	Pump A	Pump B	Pump C
Well1	1.52	1.60	1.40
Well2	1.70	1.63	1.55
Well3	1.45	1.57	1.30

	R1	R2	R3	R4	R5
Pump A	5.15	5.69	6.13	5.63	5.80
Pump B	5.12	5.47	6.05	6.12	5.71
Pump C	5.32	6.16	6.25	6.17	5.87

Formulate the problem as an LP to find the optimal solution minimizing the total cost. If you were to add well capacity which well would you add it on and how much would you be willing to invest in it?

Problem 7 (10 POINTS):

Remember the transshipment problem we discussed in class (given below):



Instead of having two different sets of variables that we had for inbound (x_{ij} variables) /outbound (y_{jk} variables) shipments (for the transshipment nodes), define the decision variables as follows:

- $\boldsymbol{X_{ijk}}$: amount shipped from supplier ¨i¨ through transshipment node ¨j¨ to demand node ¨k¨.

Please make the necessary modifications to the formulation (definition of parameters, data, constraints, objective function) to minimize the distribution cost, making sure that all relevant constraints are satisfied, and solve the problem.